



Wave power

SHATTERING FORCE

When a wave breaks, a huge weight of water topples forward with tremendous force. On some shores, most of this energy is soaked up by banks of shingle, but on rocky coasts, there is nothing to stop the full force of the breaking waves slamming into the solid rock. The tumbling water hurls loose stones at the cliffs, which weakens them and water forced into the rock can build up enough air pressure to blow the cliffs apart.

As waves pound exposed shores, they shatter and grind away solid rock, cutting it back at sea level to create caves, cliffs, and rocky reefs. Stones and sand are swept along the coast to more sheltered sites where the wave action is less violent. This allows the stones and sand to settle and form shingle banks, sandy beaches, and mudflats. Thus, in some places the coast is being carved away, while in others it is being built up.

WOW!

A big wave can hit rocks with a pressure of 7,000 lb per sq in (500 kg per sq cm)—like a car-sized hammer hitting your finger.

